

Christian Lezcano

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Data Engineer | Machine Learning Engineer

Data Engineer with a specialization in **developing scalable data infrastructure and implementing production-ready AI solutions**. My Core technical expertise lies in **Python, SQL, PostgreSQL, and AWS**, complemented by strong capabilities in **Machine Learning** utilizing frameworks like **TensorFlow and PyTorch**. I have applied these skills across diverse sectors, including beverage manufacturing, higher education, and government. My technical toolkit also includes proficiency in **ETL processes (DBT), data visualization (Power BI, MetaBase), MLOps (MLflow), containerization (Docker), CI/CD (GitHub Actions), and cloud data warehousing (Amazon Redshift)**, along with libraries like **Keras and Hugging Face**. I am seeking a challenging opportunity where I can leverage this comprehensive skill set to design robust data pipelines, build impactful predictive models, and contribute significantly to data-driven innovation within a forward-thinking organization.

Work Experience

Data Engineer at Vitaly Mineral Water

Sep 2016 to date

- Developed scalable microservices using Python and AWS (S3, RDS PostgreSQL, EC2), improving data processing and integration across multiple production facilities, resulting in a 30% reduction in data latency and enhanced system scalability.
- Built and optimized PostgreSQL databases, implementing indexing strategies and writing complex SQL queries, leading to a 40% improvement in query performance and faster data retrieval for critical business operations.
- Designed and deployed predictive models for demand forecasting and maintenance scheduling using Python and scikit-learn, achieving a 20% increase in operational efficiency through accurate regression analysis.
- Implemented CI/CD pipelines using GitHub Actions to streamline code deployment and ensure high code quality, reducing deployment time by 50% and minimizing code-related issues.
- I took ownership of our analytical data workflows. This involved integrating and preparing data from diverse sources using advanced SQL, and then transforming it into high-impact, automated reports and visualizations with Excel/Power BI, boosted by DAX and VBA for deeper insights and efficiency.

Main Technologies: Python , SQL , PostgreSQL , Microsoft Power BI , MetaBase , Microsoft Excel , Microsoft SQL Server , Extract, Transform, Load (ETL) , Data Build Tool (DBT) , Machine Learning , Apache NiFi , Docker , scikit-learn , TensorFlow , PyTorch , Keras , MLflow , Amazon Web Services (AWS), Amazon Redshift , Microsoft Excel (VBA, Power Query, Advanced Formulas), Microsoft Power BI (DAX) , Pentaho.

AI and Data Science Academic Instructor at Columbia University of Paraguay

Jul 2023 to date

- Advise undergraduate students on AI and data science projects, guiding them through the design and implementation phases, resulting in successful project completions and innovative research outcomes.

- Supervise students' progress, providing regular technical feedback and personalized support to overcome complex challenges, leading to a 20% improvement in project quality and timely submissions.
- Foster independent problem-solving and critical thinking, encouraging students to apply theoretical concepts to practical scenarios, enhancing their technical competence and confidence in AI-related projects.

Main Technologies: *Python , TensorFlow, PyTorch, Keras, scikit-learn , Pandas , NumPy , Hugging Face , Jupyter Notebooks , University Teaching , Research and Development (R&D) , Academic Research.*

Data Research Analyst at Columbia University of Paraguay

Oct 2023 to Oct 2024

Led a research-driven project to enhance academic efficiency, initially for the Computer Science program. Innovatively proposed a suite of new Key Performance Indicators (KPIs)—covering critical areas such as enrollment, retention, graduation, and academic performance—and developed a comprehensive analytical framework. This scalable system provides novel metrics and methodologies for data-driven insights into student success, designed for potential university-wide application to foster continuous improvement.

Main Technologies: *Key Performance Indicators , Research and Development (R&D) , Academic Research.*

Machine Learning & AI Projects

Financial Advisory RAG Chatbot

- Built a scalable RAG-powered chatbot leveraging NASDAQ company documents, integrating LangChain with OpenAI GPT to deliver accurate financial insights through conversational AI, improving data accessibility for non-technical users.
- Engineered an end-to-end data pipeline using PyPDF2, NLTK/spaCy, custom chunking algorithms, and FAISS indexing, enabling efficient ingestion and retrieval of thousands of financial documents.
- Optimized chatbot performance by developing a FastAPI/Uvicorn backend with Redis caching and asyncio, reducing response time latency by over 50% and ensuring real-time interactions.
- Deployed the production system with Docker and AWS S3, implementing Prometheus monitoring for system reliability and performance tracking, while achieving enhanced accuracy through hybrid BM25 + vector search with RAGAS-based validation.

Main Technologies: *Python, LangChain, OpenAI GPT, FastAPI , Docker, AWS S3 , FAISS , Redis , Prometheus , Streamlit , RAGAS*

Skills

Tech Skills: *Python , SQL , PostgreSQL , Microsoft Power BI , Microsoft Excel , Microsoft Power BI (DAX) , Microsoft SQL Server , Extract, Transform, Load (ETL), Machine Learning , Apache NiFi , TensorFlow , PyTorch , Data Build Tool (DBT) , Redis , Docker , MetaBase , Pentaho , Amazon Web Services (AWS), Amazon Redshift , scikit-learn , MLflow , TensorFlow , PyTorch , Keras , Hugging Face , LangChain, OpenAI GPT , FastAPI , FAISS , Redis , Prometheus , Streamlit , RAGAS.*

Agile Methodologies: *Scrum, Kanban*

Education

Anyone AI

Specialization in machine Learning Engineering	2025
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National University of Asuncion

M.S., Computer Science	2012
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National University of Asuncion

B.S., Computer Science Engineering	2008
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Certifications

IBM Data Science Professional Certificate

2023
